

Write a Program that accepts the no of units consumed and calculates the total bill.

• Electricity Bill Calculator

• Scenario: An electricity company charges its customers as follows:

• First 100 units: \$0.5 per unit

• Next 100 units (101-200): \$0.75 per unit

• Above 200 units: \$1 per unit

→ unit = int(input("enter units: "))

if (unit <= 0 and unit <= 100):

eb = unit \* 0.5

elif (unit >= 101 and unit <= 200):

du = unit - 100 # 125 - 100 = 25

eb = (100 \* 0.5) + (du \* 0.75)

elif (unit > 200)

du = unit - 200

eb = (100 \* 0.5) + (100 \* 0.75) + (du \* 1)

Print(f"for {unit} units the total electricity bill is {eb} \$")

o/p:- enter units: 249

for 249 units the total electricity bill is 204.0 \$

### Bit wise Operations :-

and    |    ~    ^    >>    <<    left shift  
or    not    xor    Right shift    total 5 bit

1 bit    1 bit    1 bit    1 bit    1 bit

$2^4$      $2^3$      $2^2$      $2^1$      $2^0$

16    8    4    2    1

$2^0 = 0, 1$

$2^1, 2^0 = 0, 1, 2, 3$

$2^2, 2^1, 2^0 = 0, 1, 2, \dots, 7$

Using 5 bits, I can generate upto 31

Thousand    Hundred    Ten    Unit

|   |   |   |   |            |
|---|---|---|---|------------|
| 2 | 5 | 4 | 3 |            |
| 0 | 2 | 5 | 4 | 3 - ignore |
|   | 4 | 2 | 5 | 4 - ignore |
|   |   | 2 | 5 |            |

} 2 right shift

lakh    tenthousand    Ten    Hundred    Ten    Unit

|   |   |   |   |   |   |  |
|---|---|---|---|---|---|--|
|   |   | 2 | 5 | 4 | 3 |  |
|   | 2 | 5 | 4 | 3 | 0 |  |
|   | 5 | 4 | 3 | 0 | 0 |  |
| 2 | 5 | 4 | 3 | 0 | 0 |  |

} 2 left shift

$$13 \times 2 = 26$$

|   |  |   |   |   |   |        |
|---|--|---|---|---|---|--------|
|   |  | 1 | 1 | 0 | 1 |        |
| 6 |  |   | 1 | 1 | 0 | ignore |
| 3 |  |   |   | 1 | 1 | 0      |

$$13 \times 4 = 52$$

|    |       |       |       |       |       |       |
|----|-------|-------|-------|-------|-------|-------|
|    |       |       | 1     | 1     | 0     | 1     |
| 26 |       |       | 1     | 1     | 0     | 0     |
| 52 | 1     | 1     | 0     | 1     | 0     | 0     |
|    | $2^5$ | $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
|    | 32    | 16    | 8     | 4     | 2     | 1     |

AND

|   |   |   |   |                     |
|---|---|---|---|---------------------|
| 1 | 1 | 1 | 1 | ⇒ 15                |
| 0 | 1 | 1 | 0 | ⇒ 6                 |
| 0 | 1 | 1 | 0 | ⇒ 6 - AND Operation |

15 & 6 - AND

o/p : 6

15 / 6 - or

o/p : 15



13 > 2 - right shift

8

13 < 4 - left shift

52

## Nested if Statement :

Syntax :

if (Condition): # outer if

if (Condition): # inner if

Statements of condition &

else: # inner if else

Statements of inner if else ~~statements~~

else: # outer if else

Statements of outer if else ~~statements~~

if (Cond):  
Statements

if (Cond):  
Statements

it is not nested if because  
nested if means with  
if very next statement is  
one more if like this

Exempl. for nested if

```
n = int(input("enter a num:"))
```

```
if (n >= 0):
```

```
    if (n > 0):
```

```
        print(" +ve")
```

```
    else:
```

```
        print(" Zero")
```

```
else:
```

```
    print(" -ve")
```

o/p :- enter a num : -2

-ve

When i will -2 it checks only 1st condition it is  
faster & time & space consuming compare to if else if  
ladder

```
n = int(input("enter a num :"))
```

```
if (n == 0):
```

```
    print("Zero")
```

```
elif (n > 0):
```

```
    print("pos num")
```

```
else:
```

```
    print("neg num")
```

• if...elif ladder

↳ Here if i give -2 as i/p it go through checking all conditions & go nested if is better compare to this.

---